



Beyond discharge concentrations: effects of wastewater nutrient loads on river macroinvertebrate communities

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Abstract Excessive anthropogenic inputs of nitrogen and phosphorus into surface waters remain a major challenge for river ecosystem management, with wastewater treatment plants (WWTPs) representing key point sources of nutrients, particularly in densely populated regions. In this study, we investigated the effects of nutrient loads from WWTPs of different sizes (i.e. small, medium, and large) on

macroinvertebrate communities across the Po River District (Northern Italy). Using basin-scale nutrient load estimates and macroinvertebrate metrics derived from Water Framework Directive, we assessed taxonomic, functional, and trait-based response to nitrogen and phosphorus loads. Results showed that large and small WWTPs contributed the greatest nutrient loads, and that these loads were significantly associated with shifts in community structure and functional diversity. High nutrient loads favoured tolerant taxa, reduced diversity of sensitive groups, and promoted trait modalities linked to enriched and depositional habitats. Responses of functional traits were particularly pronounced for small WWTPs, indicating strong ecological effects despite their limited size. Our findings underscore the need to integrate load-based approaches into regulatory frameworks, moving beyond concentration-based standards to capture cumulative nutrient pressures, especially under increasing hydrological intermittence and water scarcity in river networks.

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Introduction

The anthropogenic enrichment of surface waters with nitrogen (N) and phosphorus (P) is a major challenge for river management worldwide (Jwaideh et al.,

2022; Yang et al., 2025). Key nutrient sources include agriculture, livestock production, industrial activities, and discharges from wastewater treatment plants (WWTPs) (Silva et al., 2024). Urban WWTP discharges (i.e. the discharge of wastewater originating from anthropogenic activities and settlements, such as households, schools, hospitals, and other urban infrastructures) are particularly important in densely populated areas (Carey & Migliaccio, 2009) where they can release substantial nutrient load due to inefficiencies in treatment processes, excessive nutrient loads, or water scarcity in receiving water bodies. Excessive nutrient inputs contribute to eutrophication, with severe consequences for water quality, oxygen dynamic, and aquatic biodiversity (Mallin & Cahoon, 2020).

Most environmental regulations focus on nutrient concentrations in effluents (e.g. mg l^{-1}), which represent instantaneous measurements, whereas the total nutrient load (e.g. kg year^{-1}) reflects the cumulative amount of nutrients delivered over time. This distinction is crucial, as even low concentrations can result in high ecological pressure when associated with large discharge volumes or prolonged release periods. While many studies have focused on nutrient concentrations in effluents (Marino et al., 2024), the total annual load released by treatment plants into receiving water bodies can be equally or more relevant. Even treated effluents with low nutrient concentrations can produce high cumulative loads, especially during low-flow periods. The magnitude of these loads also depends on the size of the WWTP, typically expressed in population equivalent (PE), which reflect the plant's capacity to treat urban wastewaters in relation to the served population. Although environmental regulations such as the Urban Waste Water Treatment Directive (91/271/EEC) and its recent revision (Directive 2024/3019/EU) impose limits on discharge concentrations, the ecological consequences of nutrient loads for stream ecosystem functioning remain poorly understood. Moreover, these regulatory frameworks do not explicitly account for variation in discharge volume or the cumulative impact of nutrient loads on receiving ecosystems. Therefore, understanding the impacts of nutrient load on riverine community is crucial for developing sustainable river management strategies (Page et al., 2012).

WWTPs are commonly classified according to population equivalent (PE) following European

regulatory thresholds (< 2000 ; $2000\text{--}10,000$; and $> 10,000$ PE), which are widely adopted in Italy and across the EU to define treatment requirements and discharge standards. While these categories are primarily administrative, they also reflect meaningful differences in discharge volume, treatment complexity, and nutrient removal efficiency. Small plants (PE < 2000) often operate with simpler treatment infrastructure and may lack advanced nutrient removal processes, potentially resulting in higher residual loads and greater variability in effluent quality. In addition, they frequently discharge into low-order streams with limited dilution capacity, leading to stronger local ecological effects. In contrast, large plants (PE $> 10,000$) typically treat higher volumes of wastewater, often with more advanced technologies, and discharge into higher-order rivers where dilution is greater. Critically, aquatic organisms do not respond to plant size per se, but to the physical and chemical conditions, they experience. Thus, WWTP size categories are ecologically meaningful only insofar as they capture differences in discharge composition and exposure conditions.

Moreover, shifting the focus from local to basin-scale approach might help capturing the complexity of river networks as interconnected habitat patches (Loreau et al., 2003; Burgazzi et al., 2023). Similarly, seasonality might affect nutrient loads in receiving river networks (de Klein & Koelmans, 2011), highlighting the need to consider temporal variability when assessing nutrients dynamics. Consequently, a more in-depth knowledge of ecological consequences of WWTP nutrient loads is needed, to develop efficient conservation and management strategies. Benthic macroinvertebrates are among the most widely bioindicators for assessing the ecological status of rivers and streams (Sumudumali & Jayawardana, 2021). Their high taxonomic and functional diversity and differential sensitivity to environmental stressors provide detailed insights into the biological quality of water bodies (Lim & Do, 2023).

Macroinvertebrate communities respond distinctly to nutrient gradients through shifts in taxa composition, abundance, and functional structure (Prater et al., 2020; Enns et al., 2023). These responses can be quantified using multiple complementary metrics, including: (A1) taxonomic indices such as richness and abundance of

sensitive groups such as Ephemeroptera, Plecoptera, and Trichoptera (EPT); (A2) compositional metrics such as dominance and EPT/Chironomidae ratio; and (A3) functional or trait-based approaches that describe species ecological strategies including feeding habits, habitat preferences, and tolerance to pollution. Existing evidence suggests that EPT taxa decline along nutrient gradients, while tolerant groups such as Chironomidae and Oligochaeta increase, and that trait-based metrics can reveal mechanistic links between nutrient enrichment and changes in ecosystem functioning (Mor et al., 2019; Enns et al., 2023). Accordingly, these metrics are central to biomonitoring frameworks such as the Water Framework Directive (EU-WFD, 2000). Despite extensive use of macroinvertebrates for biomonitoring, few studies have explicitly linked their response to WWTP nutrient loads. Moreover, functional traits approaches, which characterise ecological roles and life-history strategy (Tachet et al., 2010), remain underused for evaluating nutrient-related impacts. Trait-based analyses can reveal mechanistic links between nutrient loads and ecosystem functioning, including processes such as organic matter decomposition and nutrient cycling (Pereda et al. 2020; Gruppuso et al., 2022).

Based on these considerations, and to better distinguish the role of cumulative nutrient inputs from WWTPs of different capacities, this study aims to: (i) evaluate whether WWTP categories, hereafter referred to as PE1 (small plants, <2000 population equivalent), PE2 (medium plants, 2000–10,000 population equivalent), and PE3 (large plants, >10,000 population equivalent), differ in their contribution to N and P loads; (ii) identify which macroinvertebrate metrics are most responsive to WWTP categories; and (iii) assess the response of functional traits to nutrient loads from WWTPs of different sizes.

We conducted this study in the Po River District (Northern Italy), a hydrologically and ecologically complex system subject to intense anthropogenic pressures. Owing to its dense population, industrialisation, and extensive wastewater infrastructure, the Po river provides an ideal setting for evaluating nutrient-induced alterations in aquatic ecosystems and for developing new metrics to better quantify the ecological effects of WWTP nutrient loads.

Materials and methods

Study area

The Po River District (Fig. 1), the largest in Italy, covers an area of approximately 86,800 km² corresponding to more than 70% of Northern Italy. The Po River collects the waters of numerous tributaries, from the Alps and the Apennines, including medium order rivers such as the Ticino, Lambro, Adda, Oglio, Mincio, and Tanaro. This complex water system crosses territories characterised by intense agricultural, industrial, and urban activities, which exert significant pressure on water quality. In 2020, there were 18,042 urban WWTP operating in Italy, 44% of which located in the Po basin (Fig. 1A). The distribution of wastewater treatment plants varied across the regions, with a greater presence along the main stem of the Po River in more densely populated and industrialised areas.

The Po District plays a central role in national water management, supporting intensive agriculture, hydropower production, and urban supply. However, its rivers often experience reduced flows during summer (Fenoglio et al., 2010; Piano et al., 2019; Burgazzi et al., 2025), limiting the dilution of treated effluents and intensifying their ecological impact. Although regional authorities have implemented river basin management plans aligned with the EU Water Framework Directive, nutrient pollution remains a persistent issue. This is especially true in low-order tributaries, where many small- to medium-sized WWTPs discharge directly into streams with limited buffering capacity.

Dataset

Data on WWTP location, population equivalent (PE), effluent discharge, total nitrogen (TN), and total phosphorus (TP) concentrations in the outflow waters of individual treatment plants were provided by Regional Environmental Protection Agencies and the Po River District Basin Authority, for the years 2017–2019. WWTPs were categorised by PE into: PE1 (<2000 PE, small), PE2 (2000–10,000 PE, medium), and PE3 (>10,000 PE, large). These thresholds follow those established by the Urban Wastewater Treatment Directive (91/271/EEC) and are widely adopted in Italy to define treatment requirements and discharge standards. The district

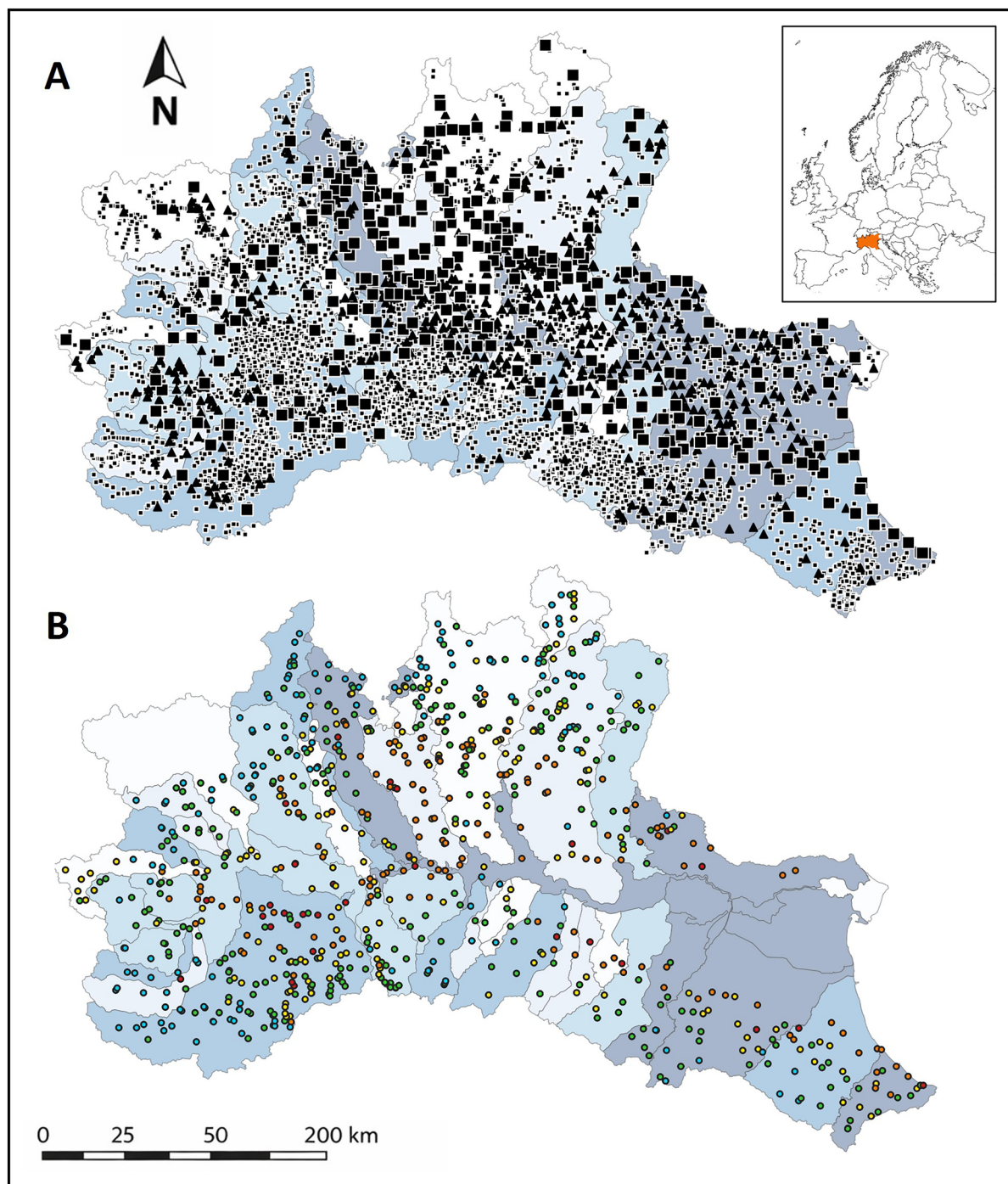


Fig. 1 Po River District maps with the basins delineated by shades from blue to white. In the top right-hand corner, the basin is indicated in orange, with its position highlighted in relation to Europe. **A** Distribution of WWTPs, categorised according to number of population equivalent. Small squares:

Category 1, Triangles; Category 2, Large squares; Category 3. **B** Macroinvertebrate sampling stations with the ecological status. Light blue= high, green= good, yellow= moderate, orange= poor, and red= bad (according to Directive 2000/60/EC)

hosts 7042 PE1, 1551 PE2, and 321 PE3 plants. Daily TN and TP loads were computed as the product of mean daily discharge and nutrient concentrations. Daily values were then aggregated and multiplied by 365 to generate annual loads (kg year^{-1}). The annual load approach was chosen because it captures the cumulative nutrient pressure exerted on receiving water bodies over a full hydrological year, in contrast with instantaneous concentration measurements that may not reflect temporal variability or low-flow periods when dilution is reduced. Annual loads were then summed by PE category (PE1, PE2, and PE3) within each of the 36 hydrographic basins, yielding category-specific basin-scale load estimates.

Macroinvertebrate data were obtained from the Environmental Protection Agency of the district and the Po River District Basin Authority (AdBPo, 2020). Communities from 36 different basins were sampled as part of the EU-WFD monitoring programme (ISPRA, 2014), with quantitative sampling conducted annually from 2017 to 2019, generally in autumn, consistent with standard Italian WFD biomonitoring protocols (Fig. 1B). Macroinvertebrate data were then matched with nutrient data by retaining only observations corresponding to basins and years for which both datasets were available. For each basin, we selected the most downstream monitoring site, ensuring coherence with cumulative nutrient load estimates. This selection criterion also ensured that no WWTPs were located downstream of the selected biological sampling point, which would, otherwise, violate the assumption that loads accumulate upstream of the monitoring location. Within each year, multiple samples may have been available for each site; therefore, values were averaged to obtain a single representative observation per site per year, and then averaged across years to minimise interannual variability. Taxa were identified to family level following the national WFD operational list. The final dataset, therefore, consisted of 36 basin-level observations, each representing the integration of temporally averaged macroinvertebrate data and corresponding aggregated nutrient loads. To harmonise the data, only plants with congruence between nutrient data and macroinvertebrate community data were considered. The final dataset included 5960 PE1, 465 PE2, and 281 PE3 plants.

Potential spatiotemporal autocorrelation was addressed at multiple levels. Spatially, by aggregating

all analyses to the basin scale ($n = 36$) and selecting a single macroinvertebrate site per basin, spatial pseudoreplication was minimised.

Macroinvertebrate metrics

Macroinvertebrate metrics were calculated at family and order levels and grouped into five commonly used bioassessment categories (Camargo et al., 2004; Poulton et al., 2015; Hering et al., 2018; Mor et al., 2019; Shortle et al., 2020): (i) High ecological status, (ii) Good, (iii) Moderate, (iv) Poor, and (v) Bad, corresponding to the standard WFD classification under Italian regulations (D.Lgs. 152/2006). Abundance indices included total abundance and group-level abundances. EPT (Ephemeroptera, Plecoptera, and Trichoptera) abundance at the family level (EPT_N) was obtained by summing individuals across all Ephemeroptera, Plecoptera, and Trichoptera families; order-level abundances for Ephemeroptera (Eph_N), Plecoptera (Ple_N), Trichoptera (Tri_N), Diptera (Dip_N), and Coleoptera (Col_N) were calculated by summing the abundances of all families belonging to each order. Richness indices included total taxa richness, EPT richness (EPT_S), and order-specific richness (Eph_S, Ple_S, Tri_S, Dip_S, and Col_S). Diversity measures consisted of proportional abundances of the major orders (Eph%, Ple%, Tri%, Dip%, and Col%) and of the family Chironomidae (Chi%), all computed from the aggregated order-level abundances relative to total sample abundance; we also quantified dominance as the share of individuals in the three most abundant taxa (DOM) and included the EPT-to-Chironomidae abundance ratio (EPT/Chi) as a composition-sensitive indicator. Functional indices captured trophic pathways and habitat conditions via functional feeding group (FFG) ratios (Cummins & Merritt, 2001)—including the production/respiration ratio (P/R), the coarse particulate organic matter to fine particulate organic matter ratio (CPOM/FPOM), the transported fine particulate organic matter to benthic fine particulate organic matter ratio (TFPOM/BFPOM), Stable Channel, and the predator-to-total taxa ratio (P/Tot)—together with family-level functional traits compiled according to Tachet et al. (2010) and applied in Mor et al. (2019) (see Supporting Information, Table S1). Biological quality was assessed using STAR_ICM (Buffagni & Furse, 2006) and the ASPT index. FFG ratios were interpreted

following Cummins & Merritt (2001). P/R ratio was defined as scrapers divided by the sum of shredders and collectors; values > 0.75 indicate autotrophy, whereas values < 0.75 reflect heterotrophy. CPOM/FPOM was calculated as shredders over collectors; values > 0.25 suggest a strong association of shredders with riparian inputs of coarse particulate organic matter (CPOM). TFPOM/BFPOM was defined as filtering collectors over gathering collectors; values > 0.50 indicate predominance of fine particulate organic matter (FPOM) in suspension, while values < 0.50 reflect greater FPOM deposition in the sediment. The Stable Channel metric was computed as (scrapers + filtering collectors)/(shredders + gathering collectors); values > 0.50 indicate stable substrates (pebbles, large boulders, or vascular plant roots), whereas values < 0.50 suggest finer and less stable substrates. We also report P/Tot, the proportion of predators, as an indicator of predation pressure within the assemblage.

Statistical analysis

All analyses were performed using R version 4.3.1 (R Core Team, 2023) with the following packages: *readxl* (Wickham et al., 2025a), *dplyr* (Wickham et al., 2023), *tibble* (Wickham & Bryan, 2023), *tidyr* (Wickham et al., 2024), *pcaPP* (Filzmoser et al., 2024), *ggplot2* (Wickham et al., 2025b), *biomonitor* (Laini et al., 2022), *FD* (Laliberté et al., 2014), *gllvm* (Niku et al., 2019), *performance* (Lüdtke et al., 2025), and *patchwork* (Pedersen, 2025).

All analyses were conducted at the basin scale ($n = 36$), using aggregated nutrient loads and corresponding macroinvertebrate metrics. To test whether WWTP size categories differed in their contribution to nutrient loads (aim A1), we fitted linear models with basin-level nutrient loads (TN and TP) as response variables and WWTP category-specific loads (PE1, PE2, and PE3) as explanatory variables. Separate models were fitted for TN and TP. This approach allowed us to explicitly assess how nutrient loads associated with each WWTP size category contributed to total basin-scale loads. Model assumptions (linearity, homoscedasticity, and normality of residuals) were evaluated using diagnostic plots (residuals vs fitted, Q-Q plots, and scale-location plots). Model performance was assessed using adjusted R^2 and residual diagnostics.

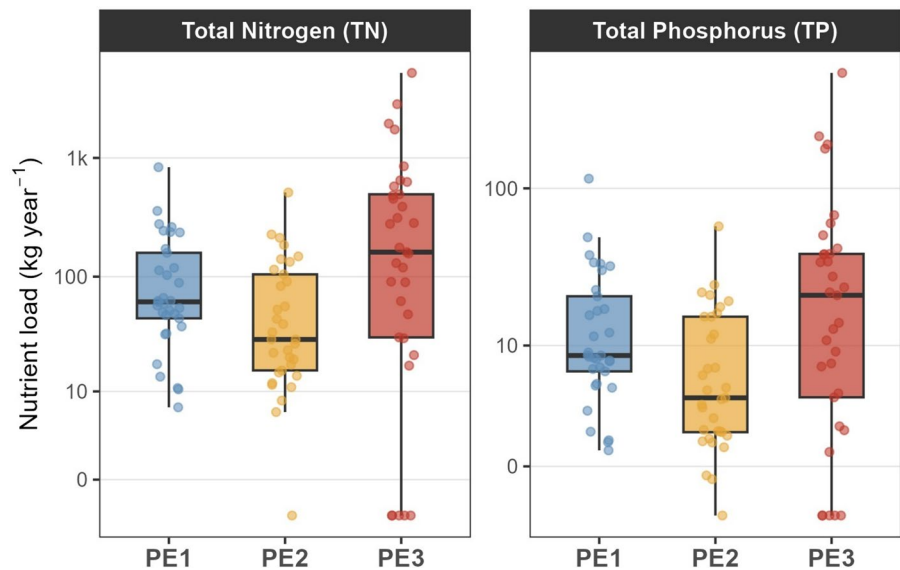
To evaluate the response of macroinvertebrate community metrics to nutrient loads (aim A2), we applied Generalised Linear Models (GLMs) using macroinvertebrate community metrics (including abundance, richness, diversity, functional indices, and biological quality indices as described in “[Macroinvertebrate metrics](#)” section) as response variables and nutrient loads (TN and TP) as explanatory variables. Because nutrient loads from different WWTP categories were strongly correlated, we reduced dimensionality using sparse Principal Component Analysis (sPCA). The sPCA generated three orthogonal components summarising variation in nutrient loads associated with PE1, PE2, and PE3 plants. These sPCA components were used as explanatory variables in GLM models. GLMs were fitted using Poisson or binomial distributions, depending on the nature of the response variable, with a log link function. Model fit was evaluated using diagnostic plots, tests for overdispersion, and pseudo- R^2 values.

To assess the effects of nutrient loads on the functional composition of macroinvertebrate communities (aim A3), expressed as community-weighted means (CWM) of biological traits (Laliberté et al., 2014), we applied multivariate generalised latent variable models (GLLVMs). The response matrix consisted of community-weighted means (CWM) of functional traits calculated at the basin scale, while explanatory variables were the same three sPCA components previously calculated representing nutrient load gradients. CWM values were derived from 47 trait modalities grouped into eight functional traits (Tachet et al., 2010; Supporting Information Table S1) (maximum body size, food, trophic status preferendum, saprobity, respiration, temperature, pH preferendum, and substrate preferendum). GLLVMs were fitted using three latent variables and assuming a Beta distribution appropriate for proportional data (Pollock et al., 2014; Niku et al., 2019). Because analyses were conducted on temporally averaged and spatially aggregated data (basin scale), potential spatiotemporal autocorrelation was reduced; however, residual spatial autocorrelation may still be present and should be considered when interpreting results.

Results

Across basins, Ephemeroptera (families: Baetidae, Caenidae, Heptageniidae, and Ephemerellidae) comprises

Fig. 2 Boxplots of total nitrogen (TN) and total phosphorus (TP) loads (kg year^{-1}) discharged by small (PE1), medium (PE2), and large (PE3) WWTP categories across the Po River District. Boxes show median and interquartile range; individual points represent single sampling sites. Y-axis is on a $\log_{10}(x+1)$ scale



60.7% of the total abundance, followed by Trichoptera (16.6%), mainly represented by Hydropsychidae, Hydroptilidae, and Rhyacophilidae, and Diptera (13.5%, with a majority of Chironomidae and Simuliidae). Plecoptera (5.5%) were primarily composed of Leuctridae, and Coleoptera (3.7%) were mostly represented by Elmidae and Dytiscidae. A complete list of taxa, including abundance and site frequency, is provided in the Supporting Information (Table S2). STAR_ICM values ranged from 0.98 to 0.12, with most sites failing to achieve good ecological status (1% high, 21% good, 30% moderate, 26% poor, and 12% bad).

Contribution of different WWTPs categories to nutrient loads

The total nutrient loads from the 5960 PE1 plants in the Po River District were 4486 and 641 t year^{-1} of N and P respectively, ranging from 7.08 to 835.97 t year^{-1} for N and from 1.50 to 114.84 t year^{-1} for P across the 36 basins. The 465 PE2 plants discharged a total of 2899 t N year^{-1} and 374 t P year^{-1} , with basin loads varying from 6.38 to 512.56 t N year^{-1} and from 0.67 to 58.12 t P year^{-1} . PE3 plants comprised 281 large facilities, which accounted for the highest outputs: 19,606 t N year^{-1} and 1792 t P year^{-1} ; among 36 basins, nitrogen loads spanned from 17.09 to 5174.29 t year^{-1} , while phosphorus ranged from 1.45 to 513.73 t year^{-1} . All reported loads refer to the

subset of WWTPs retained after data harmonisation with macroinvertebrate data (Fig. 2).

Linear model analyses highlighted that large WWTPs were the main contributor to TN and TP loads. In the model for nitrogen, the effect of PE3 was highly significant (estimate: 0.0010, $P < 0.05$), while PE1 (estimate: 0.0004, $P < 0.05$) and PE2 (estimate: 0.0002, $P < 0.05$) also had significant but smaller effects. Similarly, for phosphorus, PE3 was identified as the main contributor (estimate: 0.0010, $P < 0.05$), PE1 showed a significant effect (estimate: 0.0006, $P < 0.001$), whereas PE2 was not statistically significant (estimate: 0.0001, $P = 0.12$). sPCA clearly separated load gradients attributable to PE1, PE2, and PE3 plants (Table S3).

Community metrics influenced by the different plant categories

Twenty-eight macroinvertebrate metrics were tested against the three SPC axes. Abundance metrics showed consistent directional responses to the sPCA components. EPT_N (Fig. 3) decreased along sPC1 (estimate = -0.153) and sPC3 (-0.047) and increased along sPC2 (0.162) (Table 1).

Similarly, Eph_N decreased along sPC1 and sPC3 and increased along sPC2, whereas Ple_N increased along sPC1 and sPC3 and decreased along sPC2. Dip_N decreased along sPC1 and sPC2 and increased along sPC3. Both positive and negative responses

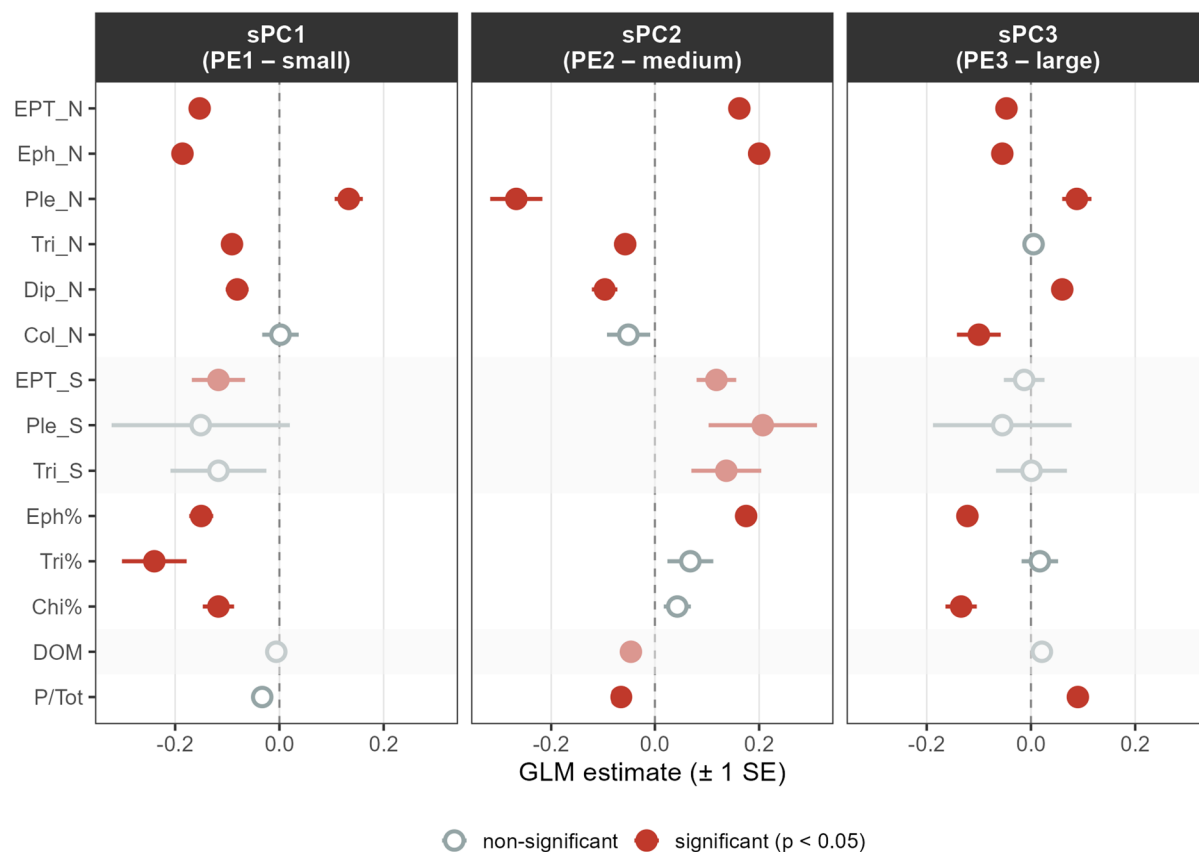


Fig. 3 Coefficient plot from generalised linear models (GLMs) fitted using macroinvertebrate metrics as response variables and sparse Principal Component Analysis axes (sPC1, sPC2, and sPC3) as explanatory variables, summarising nitrogen and phosphorus loads from small (PE1), medium

(PE2), and large (PE3) WWTPs, respectively. The plot displays model-estimated coefficients (± 1 SE) for each metric (y-axis) across the three sPCA components (panels). Metrics with coefficients significantly different from zero ($p < 0.05$) are shown in red, while non-significant associations are shown in grey

were observed across metrics, indicating that macroinvertebrate communities respond in multiple ways to nutrient load gradients, and that no single directional response captures the full ecological signal.

A significant effect of sPC1 and sPC2 was found for the abundance of Trichoptera (Tri_N), which decreased along both components, while no significant effect was observed for sPC3. Col_N showed a significant negative response only along sPC3. For richness metrics, EPT_S decreased along sPC1 and increased along sPC2, while no significant effect was detected for sPC3. Ple_S and Tri_S showed significant positive responses only along sPC2. For proportional metrics, Eph% decreased along sPC1 and sPC3 and increased along sPC2, while Tri% and Chi% showed significant responses mainly along

sPC1 and sPC3. Dominant taxa (DOM) showed a significant negative response only along sPC2, while P/Tot decreased along sPC2 and increased along sPC3. STAR_ICM and ASPT did not show significant responses (Table 1).

Functional trait responses to WWTP categories.

Trait-based responses varied strongly among WWTP categories (Fig. 4). PE1 plants (sPC1 in Fig. 4) had the higher number of trait modalities (15 out of 47), with negative effect on traits modalities associated with low saprobity and trophic state, cold water temperature, and coarse substrate preference (i.e. taxa associated to substrates such as boulders and cobbles). Positive effects were on taxa preferring muddy substrates, warmer waters, high trophic and saprobity state larger body sizes,

Table 1 Model results for response metrics, presented by metric category

Metric category	Response metric	Pseudo R ²	Comp	Estimate	Std. error	t/z value	Pr(> t)
<i>Model: glm(metric ~sPC1+SPC2+sPC3)</i>							
Abundance indices	EPT_N	0.050	sPC1	−0.153	0.006	−22.847	0.001*
			sPC2	0.162	0.004	36.585	0.001*
			sPC3	−0.047	0.005	−9.235	0.001*
	Eph_N	0.092	sPC1	−0.186	0.011	−16.87	0.001*
			sPC2	0.200	0.006	30.51	0.001*
			sPC3	−0.055	0.008	−6.95	0.001*
	Ple_N	0.010	sPC1	0.133	0.027	4.802	0.001*
			sPC2	−0.267	0.050	−5.279	0.001*
			sPC3	0.088	0.028	3.063	0.001*
	Tri_N	0.014	sPC1	−0.091	0.020	−4.459	0.001*
			sPC2	−0.057	0.021	−2.713	0.001*
			sPC3	0.005	0.016	0.371	0.711
	Dip_N	0.027	sPC1	−0.081	0.022	−3.571	0.001*
			sPC2	−0.097	0.025	−3.870	0.001*
			sPC3	0.060	0.016	3.718	0.001*
	Col_N	0.005	sPC1	0.002	0.035	0.061	0.951
			sPC2	−0.051	0.042	−1.226	0.220
			sPC3	−0.100	0.042	−2.361	0.018*
Richness indices	EPT_S	0.125	sPC1	−0.117	0.051	−2.288	0.022*
			sPC2	0.118	0.038	3.075	0.002*
			sPC3	−0.013	0.039	−0.341	0.733
	Ple_S	0.056	sPC1	−0.151	0.171	−0.886	0.375
			sPC2	0.207	0.104	1.981	0.047*
			sPC3	−0.055	0.133	−0.416	0.677
	Tri_S	0.098	sPC1	−0.117	0.092	−1.270	0.204
			sPC2	0.137	0.067	2.050	0.040*
			sPC3	0.001	0.068	0.019	0.985
Proportional abundances	Eph%	0.055	sPC1	−0.150	0.023	−6.316	0.001*
			sPC2	0.175	0.015	11.356	0.001*
			sPC3	−0.122	0.021	−5.665	0.001*
	Tri%	0.033	sPC1	−0.240	0.062	−3.841	0.001*
			sPC2	0.068	0.044	1.539	0.123
			sPC3	0.017	0.035	0.504	0.614
	Chi%	0.078	sPC1	−0.117	0.030	−3.809	0.001*
			sPC2	0.043	0.026	1.656	0.097
			sPC3	−0.134	0.030	−4.355	0.001*
Dominant taxa	DOM	0.075	sPC1	−0.006	0.017	−0.346	0.729
			sPC2	−0.046	0.020	−2.265	0.023*
			sPC3	0.021	0.016	1.308	0.191
Functional indices	P/Tot	0.014	sPC1	−0.033	0.017	−1.922	0.054
			sPC2	−0.065	0.019	−3.301	0.001*
			sPC3	0.090	0.012	7.050	0.001*

Only metrics exhibiting significance for at least one spatial principal component (SPC1, SPC2, or SPC3) are included. Significant outcomes ($P < 0.05$) are indicated by fully bolded rows. Pseudo-R² values are provided for all models

GLLVM Coefficient Plot with sPCA covariates

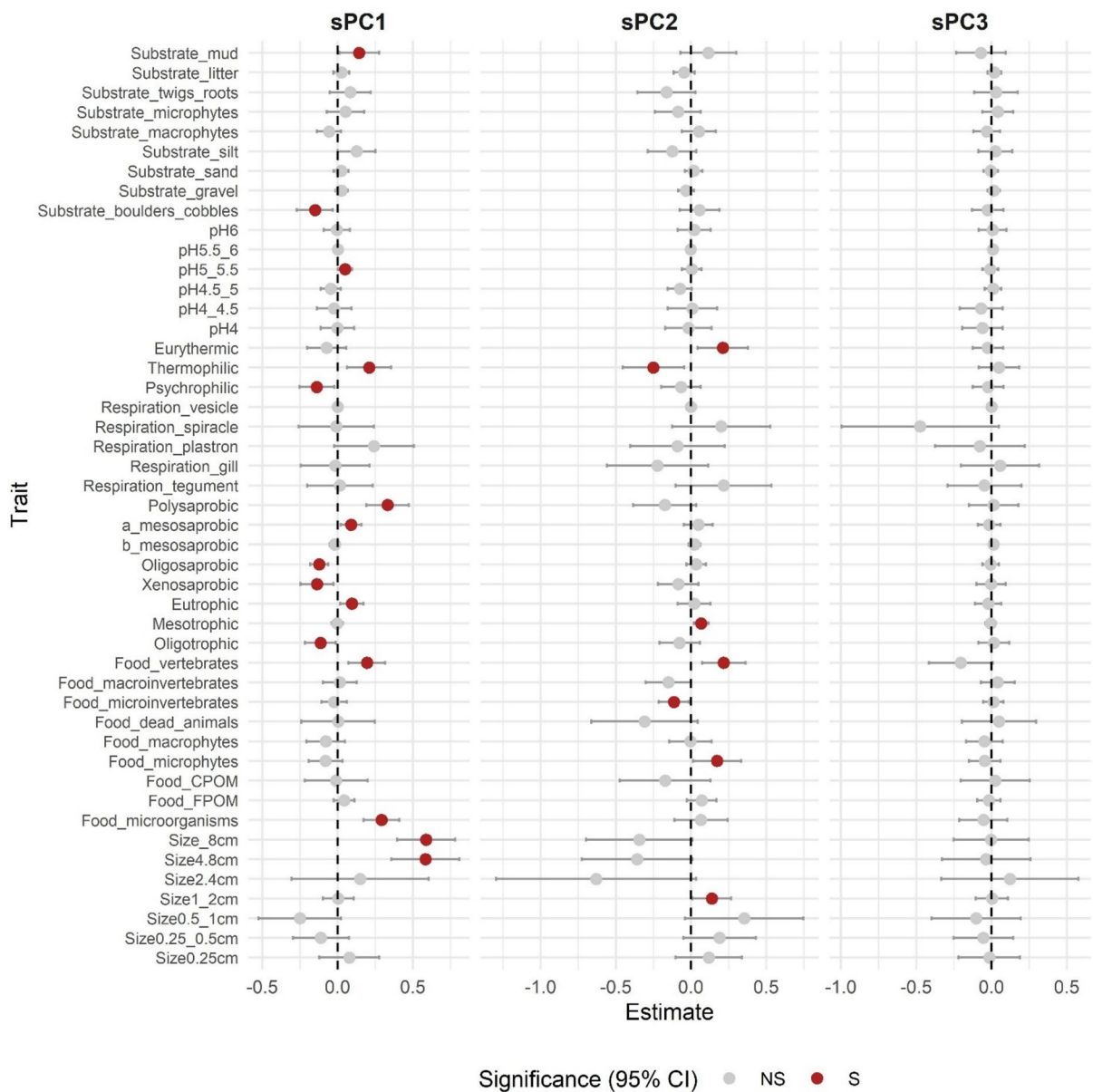


Fig. 4 Coefficient plot from the Generalised Linear Latent Variable Model (GLLVM) fitted using community-weighted mean (CWM) as response matrix and sparse Principal Component Analysis axes (sPC1, sPC2, and sPC3) as explanatory variables (summarising N and P loads from small, medium, and large WWTPs, respectively). The plot displays model-esti-

mated coefficients ($\pm 95\%$ confidence intervals) for each trait modality (y-axis) across the three sPCA components (facets). Trait modalities significantly affected by sPCA variables (confidence interval not overlapping zero) are shown in red (S), while non-significant associations are shown in grey (NS)

and feeding on *microorganisms*. PE2 plant loads (sPC2) affected seven trait modalities (Fig. 4),

positively associated mostly with *mesotrophic*, *eurythermic* and medium bodied taxa. *Thermophilic* on *microinvertebrate*-feeding taxa showed

negative associations. PE3 plant loads (sPC3) had no significant effects on functional composition.

Discussion

In this study, we investigated the differential contributions of PE1, PE2, and PE3 WWTPs to nutrient loads and their effects on macroinvertebrate metrics and functional community structure. A key conceptual point is that macroinvertebrates do not respond to plant size per se, but to the physical and chemical conditions generated by the discharged effluents. WWTP size categories are ecologically meaningful because they capture differences in discharge composition, volume, and the characteristics of receiving water bodies—not because biological communities are intrinsically sensitive to plant administrative classification. Our results highlight that PE3 plants are the main contributors to both N and P nutrient loads, consistent with their scale and catchment influence. However, the biological responses did not scale proportionally with total load, confirming that the composition and context of discharges matter as much as their magnitude. Macroinvertebrate functional traits were most strongly affected by PE1 plants, underscoring the ecological importance of discharge context—small plants discharge into low-order, poorly buffered streams where even moderate loads can cause pronounced ecological shifts.

Contribution of different WWTPs categories to nutrient loads and effects on macroinvertebrate metrics

PE3 plants (PE > 10,000) were the primary sources of N and P loads to rivers in the analysed basins, as evidenced by the linear model analysis. According to Gücker et al. (2006), discharges from large plants have higher nitrogen concentrations than small ones, as they serve densely populated areas and treat large volumes of wastewater from multiple sources. The contribution of nitrate is also higher than phosphate due to the complexity of treatment processes (Alexander et al., 2009).

The analysis of 28 metrics showed that a substantial number of macroinvertebrate indicators were significantly affected by the three sPCA components, with the strongest linked to sPC1 (PE1, small

WWTPs) and sPC3 (PE3, large WWTPs). Importantly, both positive and negative responses were observed across metrics, indicating that macroinvertebrate communities respond in multiple ways to nutrient load gradients—reflecting the complexity of ecological responses rather than a simple degradation pathway. Abundance and richness metrics of Ephemeroptera, Plecoptera, Trichoptera, Diptera, and overall EPT taxa, as well as the percentage of Ephemeroptera, were particularly responsive, confirming that WWTP discharges at both extremes of plant size exert major ecological pressures. At finer taxonomic resolution, family-level responses (e.g. Heptageniidae and Leptophlebiidae) were more pronounced than those at the order level, indicating that sensitive families may function as early-warning indicators of stress from nutrient loads. These findings are consistent with previous observations that nutrient enrichment can drive shifts in assemblage composition, with moderate loads sometimes promoting increased productivity, but excessive inputs eventually leading to declines in sensitive taxa and overall biodiversity loss (Smith et al., 1999; Carvalho et al., 2013; Hering et al., 2015). In contrast, widely applied biotic indices such as STAR_ICMi and ASPT showed limited responsiveness, underscoring the need for complementary trait-based assessments. This suggests that these multimetric indices may not fully capture gradients associated with nutrient loads alone, particularly at the basin scale, and underscores the importance of integrating trait-based and functional approaches into assessments of wastewater-related impacts. Similarly, functional ratios such as P/R, CPOM/FPOM, and TFPOM/BFPOM did not reveal consistent differences, except for CPOM/FPOM under sPC3.

These patterns indicate that macroinvertebrate responses are likely influenced by multiple interacting factors, including habitat conditions, hydrological regime, diffuse agricultural nutrient sources, and other pollutants, in addition to WWTP nutrient loads (Jones et al., 2023). Functional processes may be more strongly influenced by local habitat complexity than by nutrient enrichment per se (Petchey et al., 2015), which could explain the lack of consistent responses across sites. Taken together, these results highlight that both small and large WWTPs can exert substantial ecological impacts, that family-level metrics provide finer diagnostic power than order-level

metrics, and that the limited response of STAR_ICMi and ASPT underscores the importance of integrating trait-based and functional approaches. At the same time, these findings suggest that the observed ecological effects are not solely related to WWTP size per se, but rather to the environmental conditions generated by the discharged effluents.

Responses of macroinvertebrate functional traits to nutrient loads from different WWTP categories

Trait responses indicated substantial filtering towards tolerant and generalist taxa downstream of small WWTPs.

The first component (sPC1), representing PE1 plants (PE < 2000), had the strongest overall impact, significantly affecting 15 out of 47 tested modalities. The second component (sPC2), associated with medium-sized plants (PE 2000–10,000), influenced only seven trait modalities, reflecting more moderate nutrient enrichment and environmental filtering than category 1 plants. The third component (sPC3), summarising the effect of PE3 plants (PE > 10,000), did not show any significant influence on the functional trait composition of macroinvertebrate communities. This result may reflect several complementary processes: more diluted discharges in higher-order rivers, better treatment performance in large facilities, and the possible role of historical filtering. Communities receiving chronic high nutrient loads from large plants may have already lost sensitive taxa through local extinction, resulting in a stable but functionally impoverished community that shows limited additional response to current load variation. In contrast, communities near small WWTPs may experience more dynamic colonisation–extinction cycles, consistent with the greater number of affected trait modalities observed for sPC1. Overall, the results highlight a size-dependent impact of WWTPs on freshwater invertebrate trait composition. However, this pattern should be interpreted in the light of the environmental context in which discharges occur, rather than as a direct effect of plant size alone. While small plants appear to exert strong selective pressure towards traits adapted to enriched, unstable, and degraded habitats, medium and large plants have progressively weaker to no detectable effects. These findings underline the importance of considering plant size in ecological assessments

of WWTP impacts on aquatic communities. These results also have broader ecological implications in terms of ecosystem functioning. The decline of taxa with oligotrophic, oligosaprobic, or xenosaprobic traits—typically represented by sensitive macroinvertebrate groups such as Ephemeroptera (e.g. Heptageniidae, Leptophlebiidae, and Ephemerellidae), Plecoptera (e.g. Perlidae, Perlodidae, and Nemouridae), and several Trichoptera (e.g. Rhyacophilidae, Glossosomatidae, Polycentropodidae, and Sericostomatidae)—entails the loss of functional specialists. Other groups with similar ecological requirements include certain Elmidae (Coleoptera), freshwater mussels of Unionidae, and Neritidae (Gastropoda), all associated with clean, well-oxygenated, and structurally diverse habitats. Elmidae, for instance, are considered sensitive bioindicators of high water quality and depend on well-oxygenated riffle habitats (Elliott, 2008; González-Córdoba et al., 2020), while Unionidae are well known for their vulnerability to eutrophication and habitat degradation (Lopes-Lima et al., 2017; Bolotov et al., 2020). These organisms contribute uniquely to ecosystem processes: Shredders such as many Plecoptera and Trichoptera facilitate the breakdown of coarse organic matter; grazers such as Heptageniidae regulate algal growth; and filter-feeders (e.g. Unionidae and some Hydropsychidae) enhance water purification and nutrient retention. Their disappearance, therefore, implies not only biodiversity loss but also a reduction in essential ecological functions, ultimately leading to less resilient ecosystems. In contrast, the replacement of these sensitive taxa by groups with eutrophic and polysaprobic traits—including Oligochaeta (e.g. Naididae and Tubificidae), Chironomidae, and tolerant molluscs such as Planorbidae or Physidae—results in a functional homogenisation. The replacement of these sensitive taxa by tolerant and opportunistic groups indicates a shift towards functionally simplified communities. These taxa are opportunistic, with rapid life cycles and generalist feeding strategies, and although they can persist under enriched and unstable conditions, they offer a narrow and redundant set of functions (e.g. Bouchard, 2004). Such communities are less efficient at processing coarse organic matter, more reliant on rapid mineralisation, and less effective in sustaining ecosystem stability.

Overall, the size-dependent impact of WWTPs thus not only alters trait composition but also reshapes the functional backbone of macroinvertebrate communities: Small plants strongly promote tolerant generalists at the expense of functional specialists, with cascading effects on organic matter cycling, water purification, and ecosystem resilience.

Management implications and future perspectives

Current regulatory frameworks (91/271/EEC and Directive 2024/3019/EU) establish limits for TP and TN as maximum effluent concentrations or percentage reductions of incoming load. However, these limits do not consider the response of the receiving water body, which, in accordance with the Water Framework Directive (2000/60 EU), is required to maintain at least a good ecological status. In this study, we used two WWTP attributes most directly linked to management: plant size and the nutrient load they generate. Both were found to have significant effects on macroinvertebrate communities.

Our results underscore how the dimension of WWTPs and their associated nutrient loads influence not only the structure of communities but also their functional features. At the same time, the results indicate that nutrient loads alone do not fully explain the observed ecological patterns, highlighting the role of additional environmental drivers. More broadly, these results support the idea that management strategies should consider not only the total nutrient loads but also the composition and context of wastewater discharges. The predominance of traits associated with eutrophic and unstable conditions at impacted sites highlights the need for stricter nutrient management, particularly for category 1. These findings suggest that small-scale wastewater treatment plants (WWTPs), although limited in size, may exert relatively greater ecological pressure on receiving aquatic ecosystems compared to larger facilities. Several factors could contribute to this pattern. Smaller WWTPs often operate with simpler treatment infrastructure and may lack advanced processes for nutrient removal, potentially resulting in higher residual loads of nitrogen and phosphorus. In addition, these plants frequently discharge into low-order or headwater streams, where dilution capacity is limited and the influence of effluents is more immediate. Their operational performance may also be more variable,

particularly under conditions of hydraulic stress such as stormwater inflows, which can lead to fluctuating effluent quality. Regulatory oversight and routine monitoring are often less stringent for smaller plants, which may contribute to undetected or unmanaged impacts over time. Moreover, many small WWTPs are in relatively undisturbed or upstream catchments, where even moderate effluent inputs can cause noticeable ecological shifts. Collectively, these factors may help explain the more pronounced changes observed in the functional composition of macroinvertebrate communities downstream of smaller plants. These results highlight the potential ecological influence of small WWTPs and suggest that more targeted monitoring and management strategies could be beneficial.

In conclusion, our results highlight the importance of adopting an integrated approach to waterbody management, linking the nutrient loads to the adaptive responses of biological communities. Considering nutrients as annual loads instead of sole concentrations allows us to assess their magnitude over time and directly correlate the impact of WWTPs present in the basin on aquatic organisms. Future studies should explore the combined effects of diffuse nutrient sources, including agricultural inputs, together with habitat quality, flow regulation, and hydrological variability, to better understand how multiple interacting factors shape ecological responses to WWTP discharges and improve river ecosystem management.

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Data availability The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Conflict of interest The authors have not disclosed any conflict of interest.

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References

- AdBPo, 2020. Studio sull'origine e dinamica dei carichi inquinanti veicolati dal bacino del fiume Po e dagli altri bacini che sfociano nel mare Adriatico. https://www.adbpo.it/progetti/progetti-fsc/11_monitoraggio_a_supporto_attuazione_direttiva_2000_60_ce/studio-sullorigine-e-dinamica-dei-carichi-inquinanti-veicolati-dal-bacino-del-fiume-po-e-dagli-altri-bacini-che-sfociano-nel-mare-adriatico/
- Alexander, R. B., J. K. Böhlke, E. W. Boyer, M. B. David, J. W. Harvey, P. J. Mulholland, S. P. Seitzinger, C. R. Tobias, C. Tonitto & W. M. Wollheim, 2009. Dynamic modeling of nitrogen losses in river networks unravels the coupled effects of hydrological and biogeochemical processes. *Biogeochemistry* 93: 91–116. <https://doi.org/10.1007/s10533-008-9274-8>.
- Bolotov, I. N., A. V. Kondakov, E. S. Konopleva, I. V. Vikhrev, O. V. Aksenova, A. S. Aksenov, Y. V. Bepalaya, A. V. Borovskoy, P. P. Danilov, G. A. Dvoryankin, M. Y. Gofarov, M. B. Kabakov, O. K. Klishko, Y. S. Kolosova, A. A. Lyubas, A. P. Novoselov, D. M. Palatov, G. N. Savinov, N. M. Solomonov, V. M. Spitsyn, S. E. Sokolova, A. A. Tomilova, E. Froufe, A. E. Bogan, M. Lopes-Lima & A. A. Makhrov, 2020. Integrative taxonomy, biogeography and conservation of freshwater mussels (Unionidae) in Russia. *Sci Rep* 10: 3072. <https://doi.org/10.1038/s41598-020-59867-7>.
- Bouchard, J. W. 2004. Guide to aquatic macroinvertebrates of the upper midwest waters. *Limnology in Brazil*: 365–371.
- Buffagni, A. & M. Furse, 2006. Intercalibration and comparison —major results and conclusions from the STAR project. In Furse, M. T., D. Hering, K. Brabec, A. Buffagni, L. Sandin & P. F. M. Verdonchot (eds), *The Ecological Status of European Rivers: Evaluation and Intercalibration of Assessment Methods* Springer, Netherlands, Dordrecht: 357–364. https://doi.org/10.1007/978-1-4020-5493-8_25.
- Burgazzi, G., A. Laini, P. Viaroli, S. Fenoglio, V. C. Schreiner, R. B. Schäfer & A. Doretto, 2023. Basin-scale variables drive macroinvertebrate biomass in low-order streams across different mountain ecoregions. *Journal of Biogeography* 50: 2030–2041. <https://doi.org/10.1111/jbi.14710>.
- Burgazzi, G., A. Laini, J. England, S. Guareschi, P. Viaroli & R. Stubbington, 2025. Testing the performance of macroinvertebrate-based indices of intermittence in rivers beyond their region of development. *River Research and Applications*. <https://doi.org/10.1002/rra.4467>.
- Camargo, J. A., A. Alonso & M. De La Puente, 2004. Multimetric assessment of nutrient enrichment in impounded rivers based on benthic macroinvertebrates. *Environmental Monitoring and Assessment* 96: 233–249. <https://doi.org/10.1023/B:EMAS.0000031730.78630.75>.
- Carey, R. O. & K. W. Migliaccio, 2009. Contribution of wastewater treatment plant effluents to nutrient dynamics in aquatic systems: a review. *Environmental Management* 44: 205–217. <https://doi.org/10.1007/s00267-009-9309-5>.
- Carvalho, L., C. McDonald, C. de Hoyos, U. Mischke, G. Phillips, G. Borics, S. Poikane, B. Skjelbred, A. L. Solheim, J. Van Wichelen & A. C. Cardoso, 2013. Sustaining recreational quality of European lakes: minimizing the health risks from algal blooms through phosphorus control. *Journal of Applied Ecology* 50: 315–323. <https://doi.org/10.1111/1365-2664.12059>.
- Cummins, K. W. & R. Merritt, 2001. Application of invertebrate functional feeding groups to wetland ecosystem function and biomonitoring. *Bioassessment and Management of North American Freshwater Wetlands* John Wiley & Sons Ltd: 85–111.
- de Klein, J. J. M. & A. A. Koelmans, 2011. Quantifying seasonal export and retention of nutrients in West European lowland rivers at catchment scale. *Hydrological Processes* 25: 2102–2111. <https://doi.org/10.1002/hyp.7964>.
- Elliott, J. M., 2008. The ecology of riffle beetles (Coleoptera: Elmidae). *Freshwater Reviews* 1: 189–203. <https://doi.org/10.1608/FRJ-1.2.4>.
- Enns, D., S. Cunze, N. J. Baker, J. Oehlmann & J. Jourdan, 2023. Flushing away the future: the effects of wastewater treatment plants on aquatic invertebrates. *Water Research* 243: 120388. <https://doi.org/10.1016/j.watres.2023.120388>.
- European Union (EU), 2024. Directive (EU) 2024/3019 of the European Parliament and of the Council of 27 November 2024 concerning urban wastewater treatment (recast). Official Journal of the European Union.
- European Union (EU), 2000. Directive 2000/60/EC of the European Parliament and of the Council of 23 October 2000 establishing a framework for Community action in

- the field of water policy. *Official Journal of the European Communities* L327: 1–73.
- Fenoglio, S., T. Bo, M. Cucco, L. Mercalli & G. Malacarne, 2010. Effects of global climate change on freshwater biota: a review with special emphasis on the Italian situation. *Italian Journal of Zoology* 77: 374–383. <https://doi.org/10.1080/11250000903176497>.
- Filzmoser, P., H. Fritz, K. Kalcher & V. Todorov, 2024. pcaPP: Robust PCA by projection pursuit.
- González-Córdoba, M., M. del-Carmen-Úñiga, L.-P. Giraldo, Y.-P. Ramírez & J. Chará, 2020. Sensibilidad de Elmidae (Insecta: Coleoptera) a la perturbación del hábitat y la calidad fisicoquímica del agua en ambientes lóticos de los Andes colombianos. *Revista de Biología Tropical* 68: 601–622. <https://doi.org/10.15517/rbt.v68i2.36702>
- Gruppuso, L., A. Doretto, E. Falasco, S. Fenoglio, M. Freppaz, M. E. Benbow & F. Bona, 2022. Flow intermittency affects leaf decomposition and benthic consumer communities of Alpine streams: a case study along the Po River. *Water* 14: 258.
- Gücker, B., M. Brauns & M.T. Pusch, 2006. Effects of wastewater treatment plant discharge on ecosystem structure and function of lowland streams. *Journal of the North American Benthological Society* 25: 313–329. [https://doi.org/10.1899/0887-3593\(2006\)25%255B313:EOWTPD%255D2.0.CO;2](https://doi.org/10.1899/0887-3593(2006)25%255B313:EOWTPD%255D2.0.CO;2)
- Hering, D., L. Carvalho, C. Argillier, M. Beklioglu, A. Borja, A. C. Cardoso, H. Duel, T. Ferreira, L. Globovnik, J. Hanganu, S. Hellsten, E. Jeppesen, V. Kodeš, A. L. Solheim, T. Nöges, S. Ormerod, Y. Panagopoulos, S. Schmutz, M. Venohr & S. Birk, 2015. Managing aquatic ecosystems and water resources under multiple stress—An introduction to the MARS project. *Science of the Total Environment* 503–504: 10–21. <https://doi.org/10.1016/j.scitotenv.2014.06.106>.
- Hering, D., A. Borja, J. I. Jones, D. Pont, P. Boets, A. Bouchez, K. Bruce, S. Drakare, B. Hänfling, M. Kahlert, F. Leese, K. Meissner, P. Mergen, Y. Reyjol, P. Segurado, A. Vogler & M. Kelly, 2018. Implementation options for DNA-based identification into ecological status assessment under the European water framework directive. *Water Research* 138: 192–205. <https://doi.org/10.1016/j.watres.2018.03.003>.
- ISPRA, 2014. Linee guida per la valutazione della componente macrobentonica fluviale ai sensi del DM 260/2010.
- Jones, J. I., C. E. M. Lloyd, J. F. Murphy, A. Arnold, C. P. Duerdoh, A. Hawczak, J. L. Pretty, P. J. Johnes, J. E. Freer, M. W. Stirling, C. Richmond & A. L. Collins, 2023. What do macroinvertebrate indices measure? Stressor-specific stream macroinvertebrate indices can be confounded by other stressors. *Freshwater Biology* 68: 1330–1345. <https://doi.org/10.1111/fwb.14106>.
- Jwaideh, M. A. A., E. H. Sutanudjaja & C. Dalin, 2022. Global impacts of nitrogen and phosphorus fertiliser use for major crops on aquatic biodiversity. *International Journal of Life Cycle Assessment* 27: 1058–1080. <https://doi.org/10.1007/s11367-022-02078-1>.
- Laini, A., S. Guareschi, R. Bolpagni, G. Burgazzi, D. Bruno, C. Gutiérrez-Cánovas, R. Miranda, C. Mondy, G. Várbíró & T. Cancellario, 2022. biomonitorR: an R package for managing ecological data and calculating biomonitoring indices. *PeerJ* 10: e14183. <https://doi.org/10.7717/peerj.14183>.
- Laliberté, E., P. Legendre & B. Shipley, 2014. FD: measuring functional diversity from multiple traits, and other tools for functional ecology.
- Lim, S. & Y. Do, 2023. Macroinvertebrate conservation in river ecosystems: challenges, restoration strategies, and integrated management approaches. *Entomological Research* 53: 271–290. <https://doi.org/10.1111/1748-5967.12665>.
- Lopes-Lima, M., R. Sousa, J. Geist, D. C. Aldridge, R. Araujo, J. Bergengren, Y. Bepalaya, E. Bódis, L. Burlakova, D. Van Damme, K. Douda, E. Froufe, D. Georgiev, C. Gumpinger, A. Karatayev, Ü. Kebapçı, I. Killeen, J. Lajtner, B. M. Larsen, R. Lauceri, A. Legakis, S. Lois, S. Lundberg, E. Moorkens, G. Motte, K.-O. Nagel, P. Ondina, A. Outeiro, M. Paunovic, V. Prié, T. von Proschwitz, N. Riccardi, M. Rudzite, M. Rudzitis, C. Scheder, M. Seddon, H. Şereflişan, V. Simić, S. Sokolova, K. Stoeckl, J. Taskinen, A. Teixeira, F. Thielen, T. Trichkova, S. Varandas, H. Vicentini, K. Zajac, T. Zajac & S. Zogaris, 2017. Conservation status of freshwater mussels in Europe: state of the art and future challenges. *Biological Reviews* 92: 572–607. <https://doi.org/10.1111/brv.12244>.
- Loreau, M., N. Mouquet & R. D. Holt, 2003. Meta-ecosystems: a theoretical framework for a spatial ecosystem ecology. *Ecology Letters* 6: 673–679. <https://doi.org/10.1046/j.1461-0248.2003.00483.x>.
- Lüdecke, D., D. Makowski, M.S. Ben-Shachar, I. Patil, P. Waggoner, B.M. Wiernik, R. Thériault, V. Arel-Bundock, M. Jullum, B. gjo11, E. Bacher & J. Luchman, 2025. Performance: assessment of regression models performance.
- Mallin, M. A. & L. B. Cahoon, 2020. The hidden impacts of phosphorus pollution to streams and rivers. *BioScience* 70: 315–329. <https://doi.org/10.1093/biosci/biaa001>.
- Marino, A., S. Bertolotti, M. Macri, F. Bona, S. Bonetta, E. Falasco, M. Minella & S. Fenoglio, 2024. Impact of wastewater treatment and drought in an Alpine region: a multidisciplinary case study. *Heliyon* 10: e35290. <https://doi.org/10.1016/j.heliyon.2024.e35290>.
- Mor, J.-R., S. Dolédec, V. Acuña, S. Sabater & I. Muñoz, 2019. Invertebrate community responses to urban wastewater effluent pollution under different hydro-morphological conditions. *Environmental Pollution* 252: 483–492. <https://doi.org/10.1016/j.envpol.2019.05.114>.
- Niku, J., F. K. C. Hui, S. Taskinen & D. I. Warton, 2019. gllvm: fast analysis of multivariate abundance data with generalized linear latent variable models in R. *Methods in Ecology and Evolution* 10: 2173–2182. <https://doi.org/10.1111/2041-210X.13303>.
- Page, T., A. L. Heathwaite, B. Moss, C. Reynolds, K. J. Beven, L. Pope & R. Willows, 2012. Managing the impacts of nutrient enrichment on river systems: dealing with complex uncertainties in risk analyses. *Freshwater Biology* 57: 108–123. <https://doi.org/10.1111/j.1365-2427.2012.02756.x>.
- Pedersen, T.L., 2025. patchwork: the composer of plots.
- Pereda, O., L. Solagaistua, M. Atristain, I. de Guzmán, A. Larrañaga, D. von Schiller & A. Eloşegi, 2020. Impact of wastewater effluent pollution on stream functioning: a whole-ecosystem manipulation experiment.

- Environmental Pollution 258: 113719. <https://doi.org/10.1016/j.envpol.2019.113719>.
- Petchey, O. L., M. P. J. Montarp, T. M. Massie, S. Kéfi, A. Ozgul, M. Weilenmann, G. M. Palamara, F. Altermatt, B. Matthews, J. M. Levine, D. Z. Childs, B. J. McGill, M. E. Schaepman, B. Schmid, P. Spaak, A. P. Beckerman, F. Pennekamp & I. S. Pearse, 2015. The ecological forecast horizon, and examples of its uses and determinants. *Ecology Letters* 18: 597–611. <https://doi.org/10.1111/ele.12443>.
- Piano, E., A. Doretto, E. Falasco, S. Fenoglio, L. Gruppato, D. Nizzoli, P. Viaroli & F. Bona, 2019. If Alpine streams run dry: the drought memory of benthic communities. *Aquat Sci* 81: 32. <https://doi.org/10.1007/s00027-019-0629-0>.
- Pollock, L. J., R. Tingley, W. K. Morris, N. Golding, R. B. O'Hara, K. M. Parris, P. A. Vesk & M. A. McCarthy, 2014. Understanding co-occurrence by modelling species simultaneously with a joint species distribution model (JSDM). *Methods in Ecology and Evolution* 5: 397–406. <https://doi.org/10.1111/2041-210X.12180>.
- Poulton, B. C., J. L. Graham, T. J. Rasmussen & M. L. Stone, 2015. Responses of macroinvertebrate community metrics to a wastewater discharge in the upper blue river of Kansas and Missouri, USA. *Journal of Water Resource and Protection* 7: 1195–1220. <https://doi.org/10.4236/jwarp.2015.715098>.
- Prater, C., P. M. Bumpers, L. M. Demi, A. D. Rosemond & P. D. Jeyasingh, 2020. Differential responses of macroinvertebrate ionomes across experimental N:P gradients in detritus-based headwater streams. *Oecologia* 193: 981–993. <https://doi.org/10.1007/s00442-020-04720-x>.
- R Core Team, 2023. R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria. <https://www.R-project.org/>
- Shortle, J. S., J. R. Mihelcic, Q. Zhang & M. Arabi, 2020. Nutrient control in water bodies: a systems approach. *Journal of Environmental Quality* 49: 517–533. <https://doi.org/10.1002/jeq2.20022>.
- Silva, G. P., A. F. Souza, M. V. Santana & C. Moulton, 2024. Influence of wastewater effluent on trophic structure and stability of a neotropical stream food web. *Ecological Indicators* 161: 111974. <https://doi.org/10.1016/j.ecolind.2024.111974>.
- Smith, V. H., G. D. Tilman & J. C. Nekola, 1999. Eutrophication: impacts of excess nutrient inputs on freshwater, marine, and terrestrial ecosystems. *Environmental Pollution* 100: 179–196. [https://doi.org/10.1016/S0269-7491\(99\)00091-3](https://doi.org/10.1016/S0269-7491(99)00091-3).
- Sumudumali, R. G. I. & J. M. C. K. Jayawardana, 2021. A review of biological monitoring of aquatic ecosystems approaches: with special reference to macroinvertebrates and pesticide pollution. *Environmental Management* 67: 263–276. <https://doi.org/10.1007/s00267-020-01423-0>.
- Tachet, H., P. Richoux, M. Bournaud & P. Usseglio-Polatera, 2010. Invertébrés d'eau douce - systématique, biologie, écologie - CNRS Editions, CNRS EDITIONS, Paris:
- Wickham, H. & J. Bryan, 2023. tibble: simple data frames.
- Wickham, H., R. François, L. Henry & K. Müller, 2023. dplyr: a grammar of data manipulation.
- Wickham, H., D. Vaughan, M. Girlich, M. Ushey, K. 2024. tidy: tidy messy data.
- Wickham, H., J. Bryan, P. Posit, M. Kalicinski, K. Valery, C. Leittenne, B. Colbert, D. Hoerl, E. Miller E, 2025a. readxl: read excel files.
- Wickham, H., W. Chang, L. Henry, T.L. Pedersen, K. Takahashi, C. Wilke, K. Woo, H. Yutani, D. Dunnington, T. Brand, P.B.C. van den Posit, 2025b. ggplot2: create elegant data visualisations using the grammar of graphics.
- Yang, J., Z. Li, E. Jeppesen, D. A. Moi, Y. Liu, Y. Mo, X. Bai, F. Wang, Z. Xie & J. Zhang, 2025. Urbanization alters river multifunctionality by reducing macroinvertebrate diversity in highly human-impacted plain river networks. *Limnology and Oceanography*. <https://doi.org/10.1002/lno.70221>.

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